STAT 4830 - SPRING 2026

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Optimizing Hyperparameter Tuning Methods for *Evolutionary Strategies*

TABLE OF CONTENTS

1. Problem Statement
2. Technical Approach
3. Results
4. Limitations
5. Next Steps

01

Problem Statement

Can we fine-tune LLMs without gradients?

- **Problem:** $max_\theta R(\theta)$
- **Zeroth-order Idea**
  - Estimate the gradient of *R* using perturbations
  - Forward passes only
  - Parameter-space exploration (vs. RLHF's action space exploration)
- **Why zeroth order**
  - No backward pass → low memory
  - Massive parallelization (scalability)
  - Works with discrete / black-box rewards
- **The Challenge**
  - Variance scales with dimension
  - Forward passes are expensive
  - High-d parameter space
- **Key question**
  - Can estimator design make Evolution Strategies (ES) viable at LLM scale?

$$\theta$$

$$\downarrow$$

$$LLM(\theta)$$

$$\downarrow$$

$$Reward\ R$$

$$\nabla R(\theta) \approx R(\theta + \sigma u)$$

02

Setup & Context

How has ES been used and where is it now?

- **Pre-2017**
  - Evolution Strategies was originated from genetic algorithms in 1960s-70s as a black-box optimization
  - CMA-ES (2001) and Natural ES (2014) in the 21st century
  - ES/ZO as stochastic gradient estimators via random perturbations
- **2017 by OpenAI**
  - ES becomes “modern” in ML by scaling massively on RL benchmarks (Atari/MuJoCo)
  - Showed zeroth-order methods like ES can match complex RL techniques
- **Post-2017**
  - RLHF (RL from Human Feedback) became a dominant LLM fine-tuning method
  - ES was recognized for its viability in (1) black-box models (2) memory bottlenecks (3) non-differentiable/discrete rewards
- **2025-2026**
  - Ongoing discussions about ES in LLM fine-tuning
  - Extensions: LoRA/subspace/quantized ES
  - Caution: emerging evidence of different failure modes (e.g., catastrophic forgetting risk)

Source: Hansen, N., & Ostermeier, A. (2001). Completely derandomized self-adaptation in evolution strategies. Evolutionary Computation, 9(2), 159–195; Wierstra, D., et al. (2014). Natural evolution strategies. Journal of Machine Learning Research, 15(1), 949–980.; Salimans, T., et al. (2017). Evolution strategies as a scalable alternative to reinforcement learning. *arXiv:1703.03864*; Mania, H., et al. (2018). Simple random search of static linear policies is competitive for reinforcement learning. *NeurIPS*; Qiu et al. (2025). Evolution strategies at scale: LLM fine-tuning beyond reinforcement learning. *arXiv:2509.24372*.

03

First Attempts

Week 1 Progress and Lessons

- **Goal:** Compare the ES and PG (Policy Gradient; first-order method baseline) in an LLM fine-tuning under simple environment (Wordle)
- **What happened:**
  - No weight-level perturbation access in Prime Intellect
    - Should have hosted model locally (i.e., Hugging Face, vLLM) and connected to Prime Intellect GPU but didn't have credits back then
  - Switched to modular prompt optimization
    - Divided prompt into modules, one-hot encoded into a vector, applied ES there
    - Prompt space was **discrete and non-smooth**, meaning the result wasn't significant
- **Lesson:** Zeroth-order requires continuous parameter space + stable reward evaluation.

04

Technical Approaches

Our Pivot - ARS Exploration

- **Motivation**
  - Fundamentally understand how evolutionary strategy algorithms work before asking the following question:
  - How can we optimize ES algorithms and tuning methods for them to efficiently find hyperparameters?
- **Implementation**
  - Compare and contrast the performance of Vanilla-ES and ARS on LQR and Pendulum

Technical Approaches - ES

Algorithm 1 Basic ES Algorithm

Require: Pretrained LLM with initial parameters $\boldsymbol{\theta}_0$, reward function $R(\cdot)$, total iterations T, population size N, noise scale σ, learning rate α.

for $t = 1$ to T **do** ▷ outer ES iterations
  for $n = 1$ to N **do**
    Sample noise $\boldsymbol{\varepsilon}_n \sim \mathcal{N}(0, \boldsymbol{I})$
    Compute reward for perturbed parameters: $R_n = R(\boldsymbol{\theta}_{t-1} + \sigma \cdot \boldsymbol{\varepsilon}_n)$
  end for
  Normalize R_n
  Update model parameters as $\boldsymbol{\theta}_t \leftarrow \boldsymbol{\theta}_{t-1} + \alpha \cdot \frac{1}{N} \sum_{n=1}^{N} R_n \boldsymbol{\varepsilon}_n$
end for

Sample Perturbations

Evaluate Perturbed Models

Update Parameters

Qiu, Xin, et al. "Evolution Strategies at Scale: LLM Fine-Tuning Beyond Reinforcement Learning." ArXiv.org, Sept. 2025, https://doi.org/10.48550/arxiv.2509.24372.

Technical Approaches - ARS

Algorithm 1 Augmented Random Search (ARS): four versions **V1**, **V1-t**, **V2** and **V2-t**

1: **Hyperparameters:** step-size α, number of directions sampled per iteration N, standard deviation of the exploration noise ν, number of top-performing directions to use b ($b < N$ is allowed only for **V1-t** and **V2-t**)
2: **Initialize:** $M_0 = \mathbf{0} \in \mathbb{R}^{p \times n}$, $\mu_0 = \mathbf{0} \in \mathbb{R}^n$, and $\Sigma_0 = \mathbf{I}_n \in \mathbb{R}^{n \times n}$, $j = 0$.
3: **while** ending condition not satisfied **do**
4: Sample $\delta_1, \delta_2, \ldots, \delta_N$ in $\mathbb{R}^{p \times n}$ with i.i.d. standard normal entries.
5: Collect $2N$ rollouts of horizon H and their corresponding rewards using the $2N$ policies

$$\textbf{V1:} \begin{cases} \pi_{j,k,+}(x) = (M_j + \nu\delta_k)x \\ \pi_{j,k,-}(x) = (M_j - \nu\delta_k)x \end{cases}$$

$$\textbf{V2:} \begin{cases} \pi_{j,k,+}(x) = (M_j + \nu\delta_k)\,\mathrm{diag}\,(\Sigma_j)^{-1/2}\,(x - \mu_j) \\ \pi_{j,k,-}(x) = (M_j - \nu\delta_k)\,\mathrm{diag}(\Sigma_j)^{-1/2}(x - \mu_j) \end{cases}$$

for $k \in \{1, 2, \ldots, N\}$.
6: **V1-t, V2-t:** Sort the directions δ_k by $\max\{r(\pi_{j,k,+}), r(\pi_{j,k,-})\}$, denote by $\delta_{(k)}$ the k-th largest direction, and by $\pi_{j,(k),+}$ and $\pi_{j,(k),-}$ the corresponding policies.
7: Make the update step:

$$M_{j+1} = M_j + \frac{\alpha}{b\sigma_R} \sum_{k=1}^{b} \left[r(\pi_{j,(k),+}) - r(\pi_{j,(k),-}) \right] \delta_{(k)},$$

where σ_R is the standard deviation of the $2b$ rewards used in the update step.
8: **V2:** Set μ_{j+1}, Σ_{j+1} to be the mean and covariance of the $2NH(j+1)$ states encountered from the start of training.[1]
9: $j \leftarrow j + 1$
10: **end while**

.

Sample Perturbations

Evaluate Perturbed Models

Update Parameters

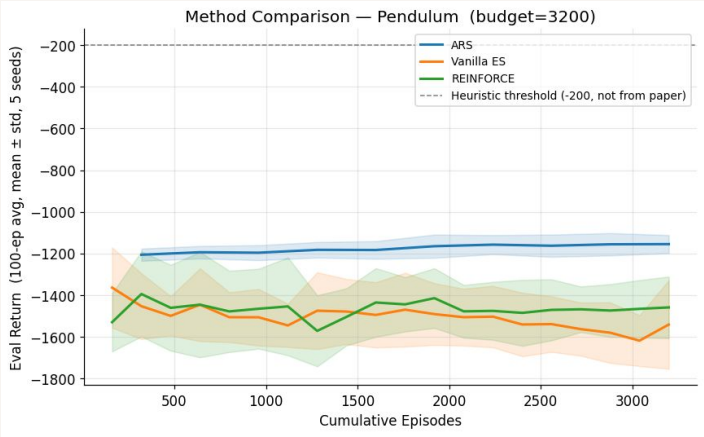
Mania, Horia, et al. "Simple Random Search of Static Linear Policies Is Competitive for Reinforcement Learning." Neural Information Processing Systems, vol. 31, Jan. 2018, pp. 1800–09. papers.nips.cc/paper/7451-simple-random-search-of-static-linear-policies-is-competitive-for-reinforcement-learning.pdf.

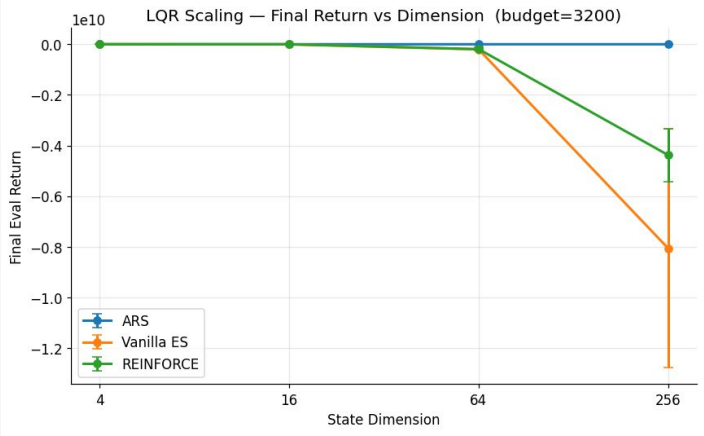
05

Initial Results

Results

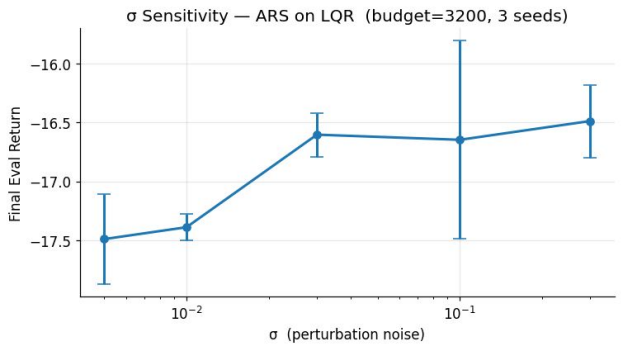
- **Core Performance and Methods Comparison**
  - ARS outperforms baselines and produces a rapid convergence for the LQR environment
  - Benefits stem from reduction of noise through symmetric perturbation, sampling top performing directions
- **Dimension Scaling**
  - In higher dimensions, ARS doesn't collapse while Vanilla ES and REINFORCE collapses exponentially

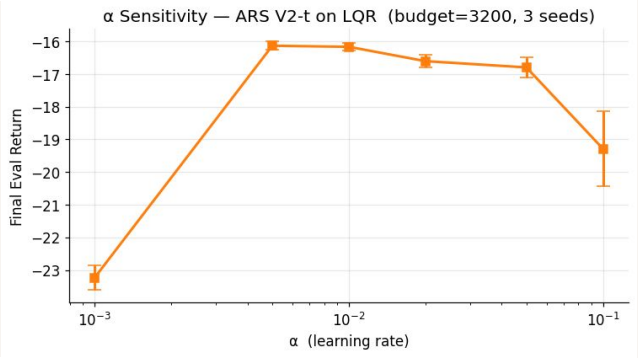


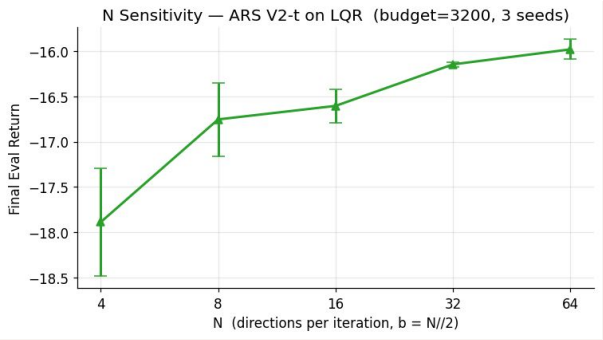


Results

- **Hyperparameter Sensitivity**
  - ARS remained stable across a full decade [0.03, 0.3] of noise standard deviation σ. Meaningful degradations only occur when perturbations are too small
  - There is a tight bound on the optimal range [0.005, 0.01] for the learning rate α
  - Performance improve monotonically as population N increases, but experiences strong diminishing returns







06

Next Steps

What's next: From Benchmarking to Mechanism-Level Understanding

Short-term (3rd deliverable)

- **ES vs ARS in identical LLM setup**
  - Reward vs forward-pass budget
  - Stability across seeds
  - Population size scaling law
- **Unified minimal ZO baseline**
  - One-sided vs two-sided
  - Rademacher vs Gaussian
  - Normalized vs unnormalized updates

Long-term goals (4th deliverable and beyond)

- **Systematic dissection of SOTA ZO/ES mechanisms**
  - MeZO, HiZOO, ESSA, BSZO, FZOO, Sparse MeZO, ...
  - Identify which components provide real gains in LLM fine-tuning: Curvature? Subspace restriction? Multi-direction fusion? Sparse updates?
- **Theory (if time permits)**
  - Variance scaling in high-dimensional LLM parameterizations
  - Population size vs effective dimension